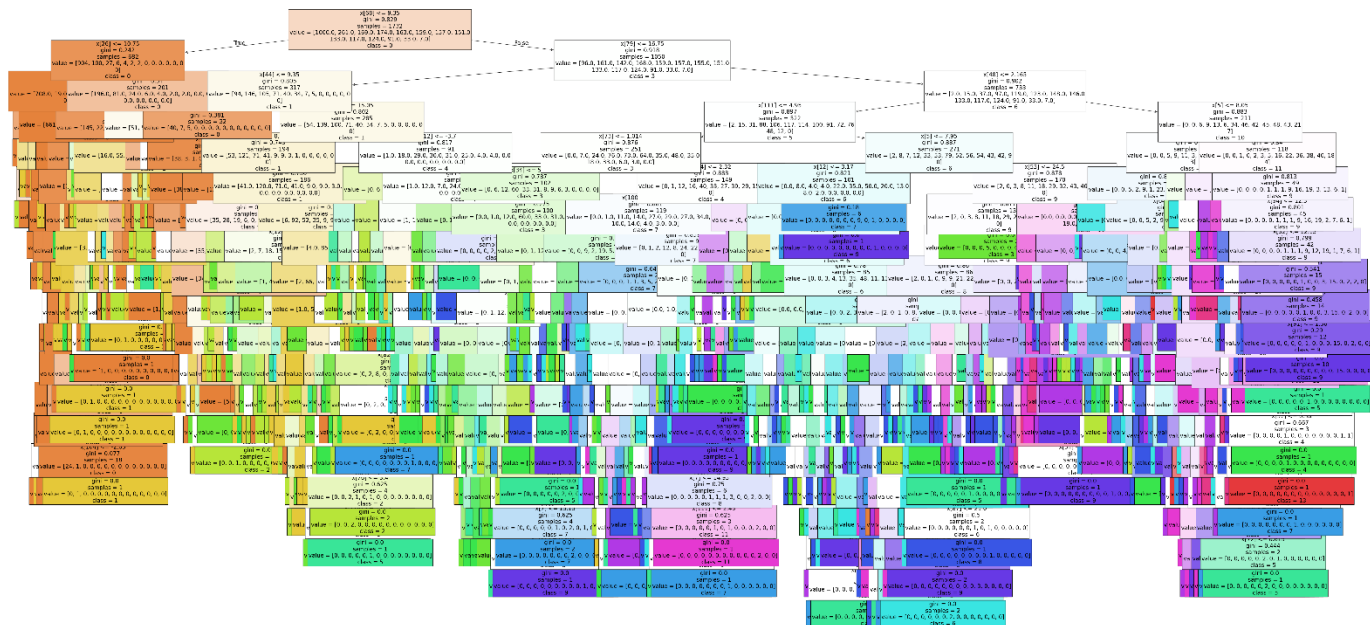


Task 2.3

Random Forest and Feature Importance Analysis of European Weather Data

1. All Weather Stations and Pleasant Weather Answers, 01.01.2010 to 31.12.2019

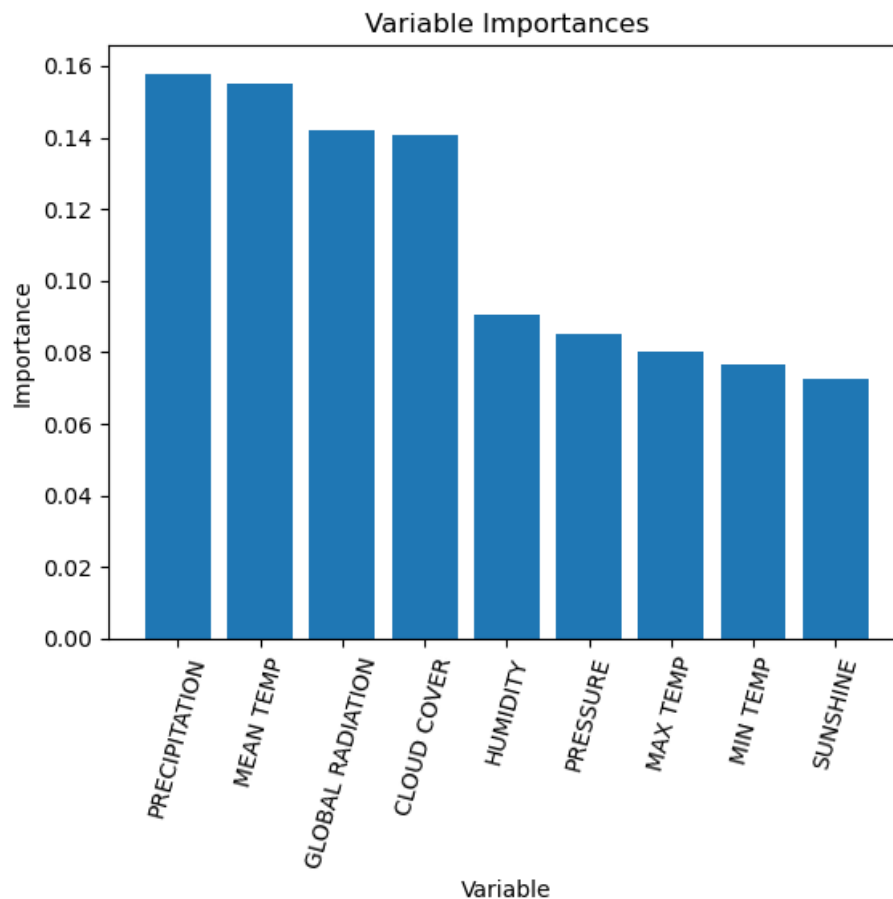


With $n_estimators = 100$, this model performed with 55.1% accuracy.

Feature Importances

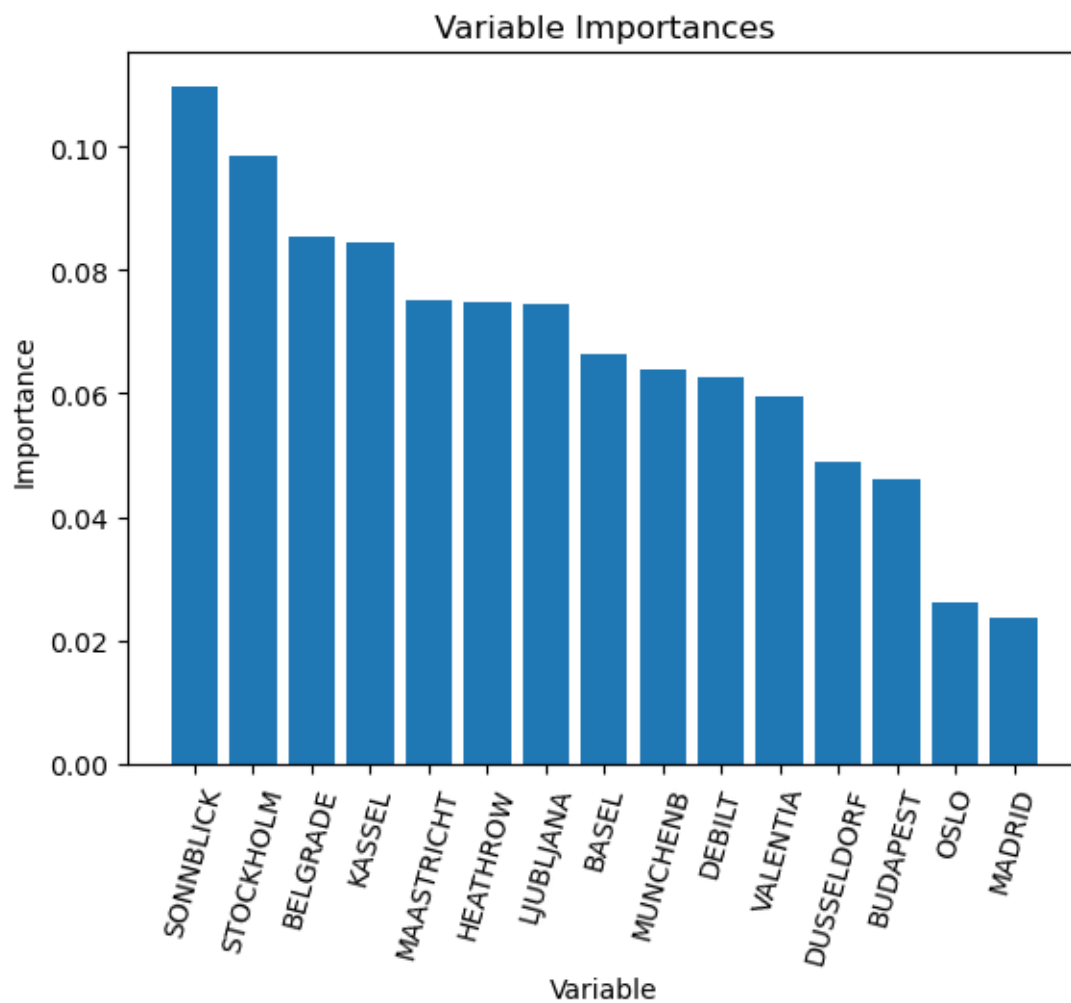
Determining which features – which of the 9 weather conditions and which of the 15 weather stations – has the most influence on the random forest model's classification of data.

Importance of Weather Conditions



When classifying data from all weather stations between 2010 and 2020, **precipitation** and **mean temperature** had the most influence on the classification performed by the model, followed by **global radiation levels** and **cloud cover**.

Importance of Individual Weather Stations

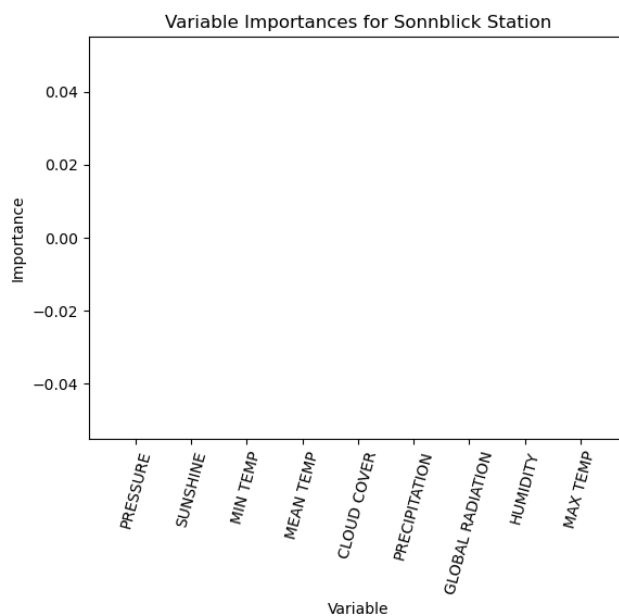


When classifying data from all weather stations between 2010 and 2020, the stations of **Sonnblick**, **Stockholm** and **Belgrade** had the most influence on the classification performed by the model.

2. Weather Data and Pleasant Weather Answers from Sonnblick, 1960-2022

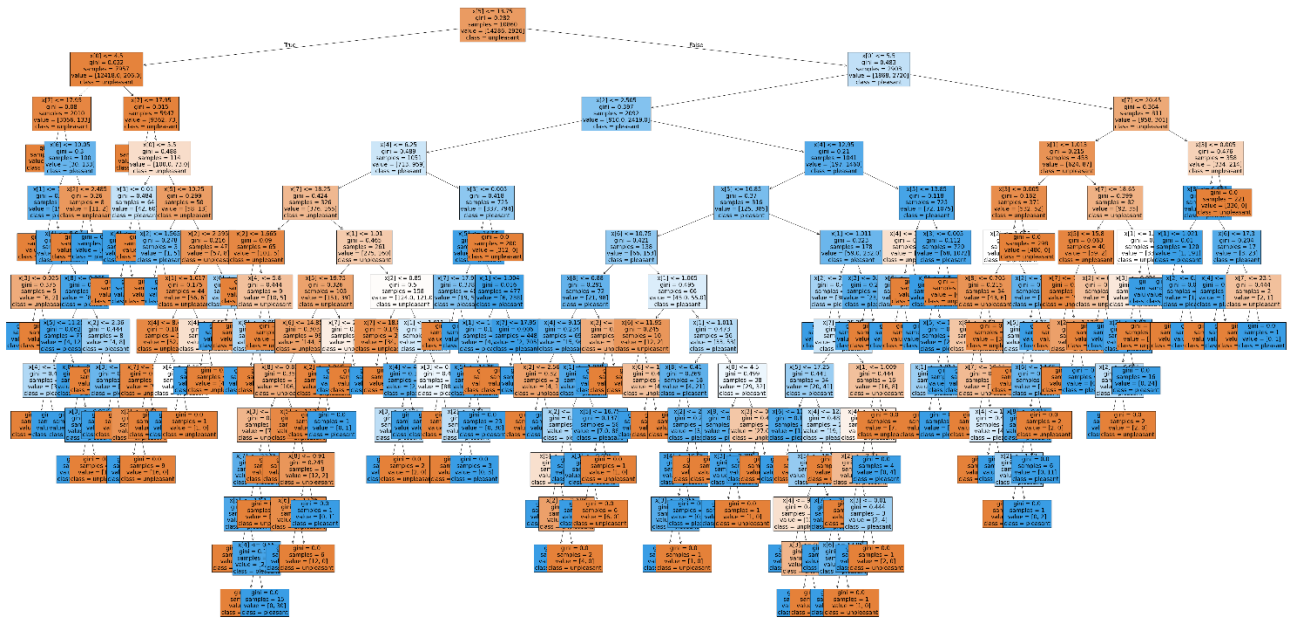
gini = 0.0 samples = 10887 value = 1.0

Feature Importance for Sonnblick Data



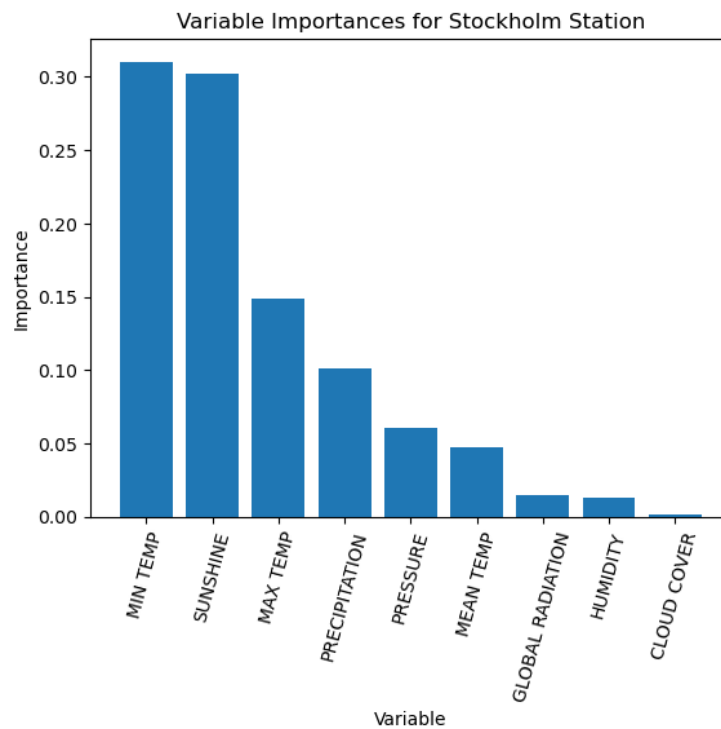
These unusual results are due to the **pleasant weather data for Sonnblick containing only negative answers.**

3. Weather Data and Pleasant Weather Answers from Stockholm, 1960-2022



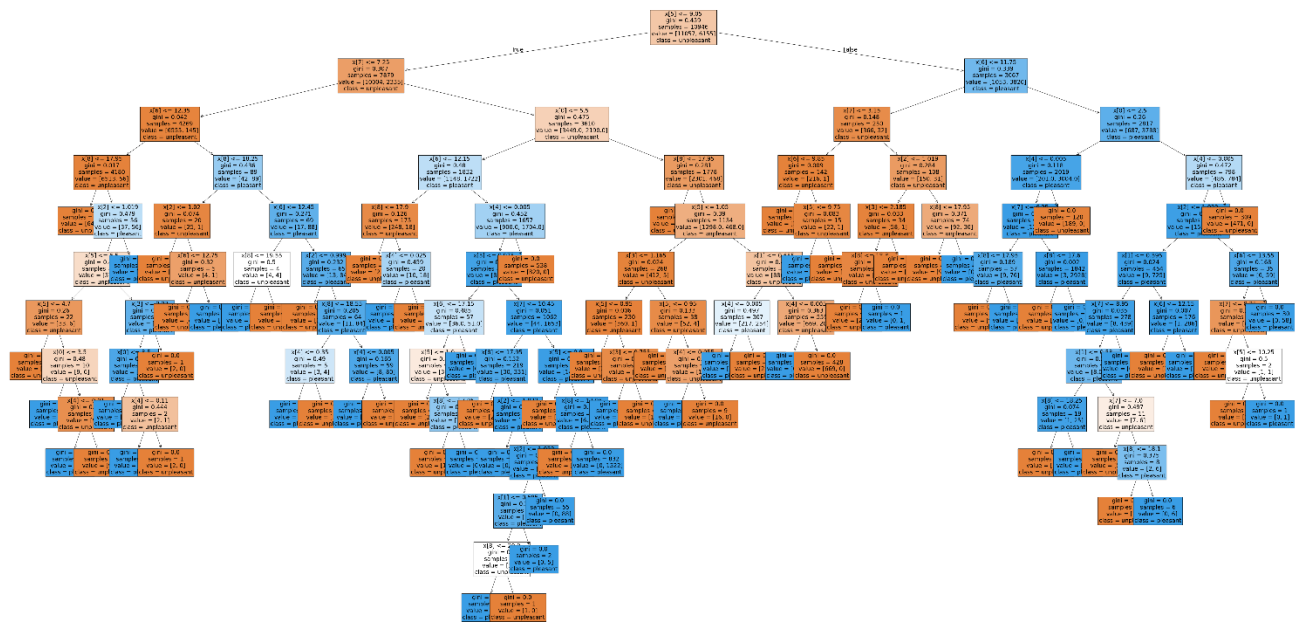
With $n_classifiers = 100$, this model showed a 100% accuracy rate.

Feature Importance for Stockholm Station Data



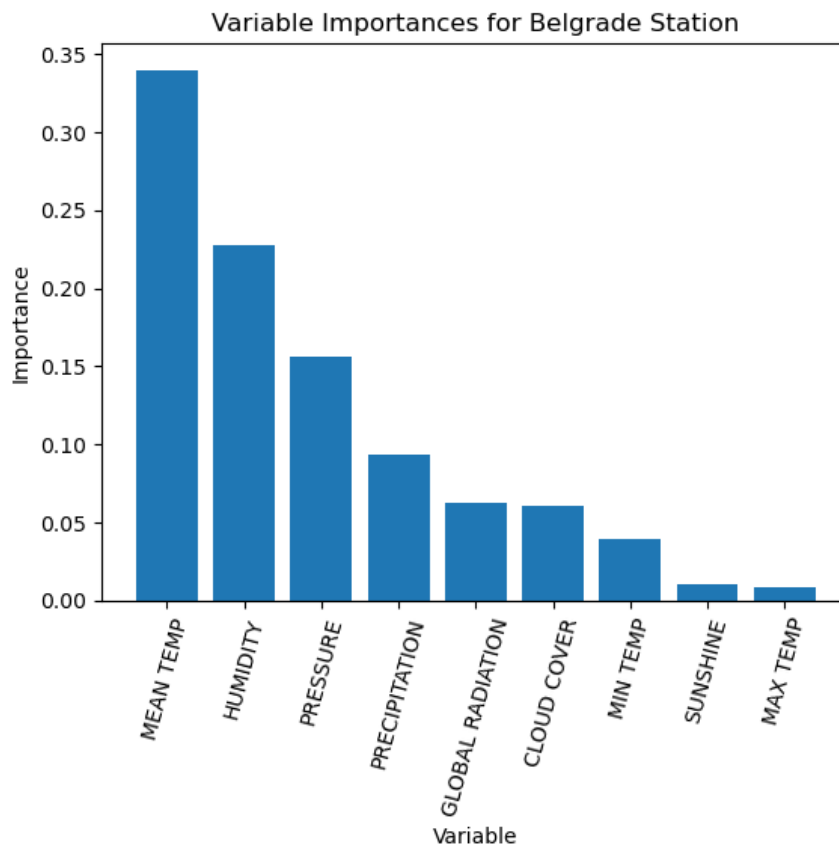
For Stockholm, **minimum temperature** and **sunshine levels** had the most influence on the random forest's classification of the weather data.

3. Belgrade Weather Data and Pleasant Weather Answers, 1960-2022



With $n_{\text{estimators}} = 100$, this model showed 100% accuracy

Feature Importance for Belgrade Weather Data

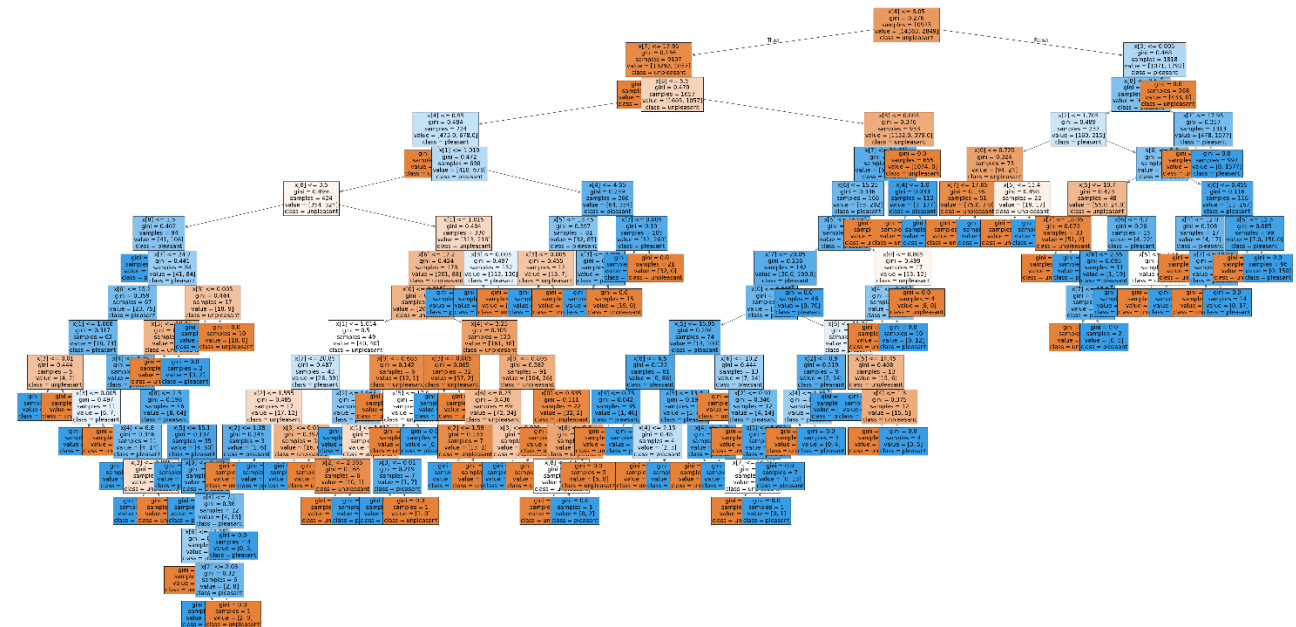


For Belgrade weather data across all years, **mean temperature** and **humidity** had the most influence on the classifications made by the random forest model.

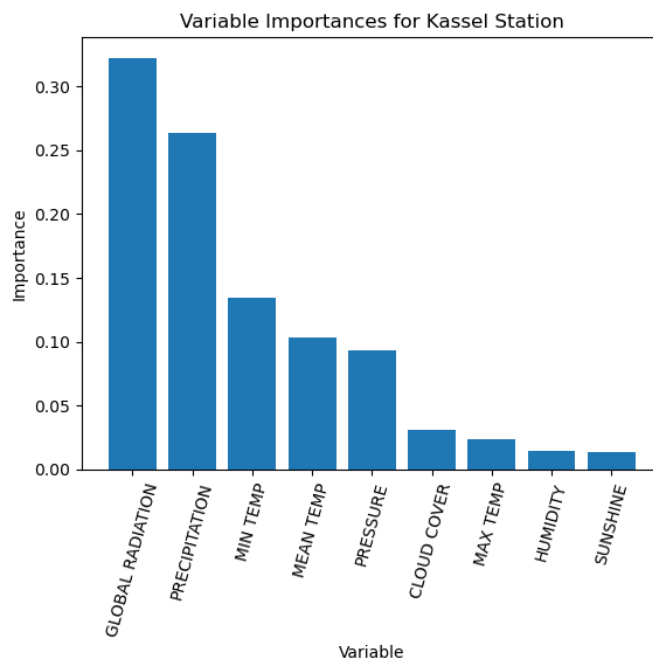
Because of the unusual nature of the Sonnblick data, I decided to test the station with the next most importance after Belgrade, Kassel.

Kassel Pleasant Weather Classifier

(n_classifiers = 100, accuracy = 1.0)



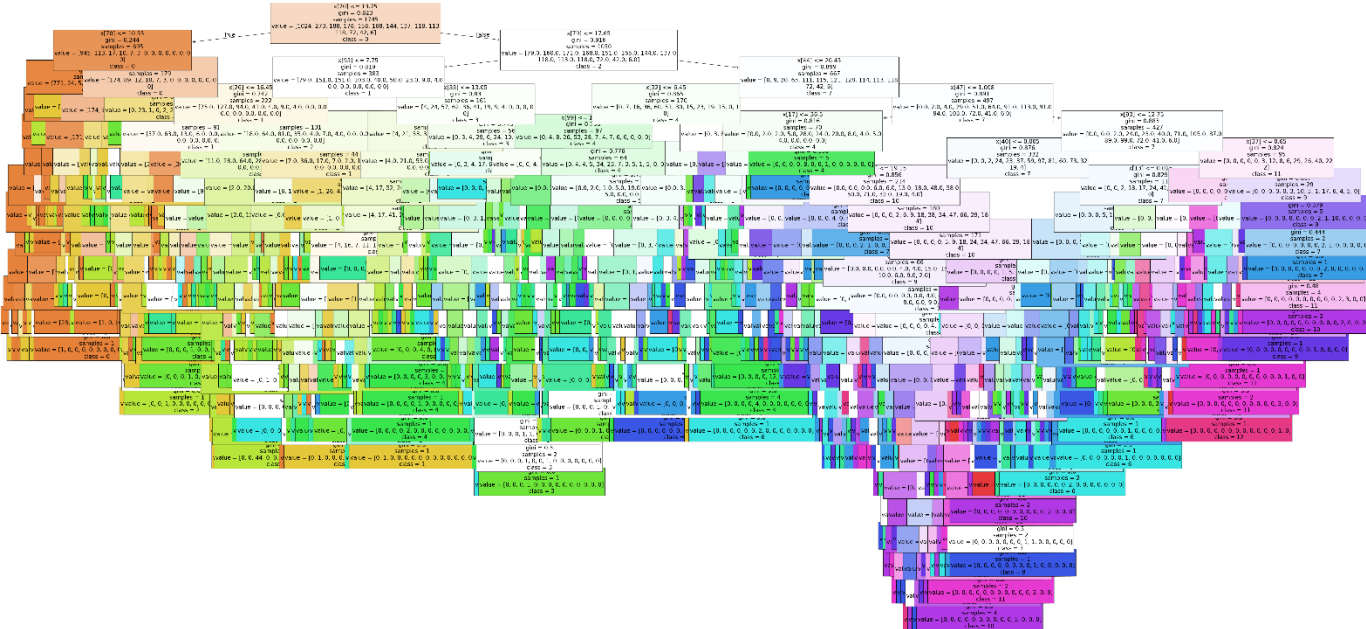
The accuracy of the model is also 100% here.



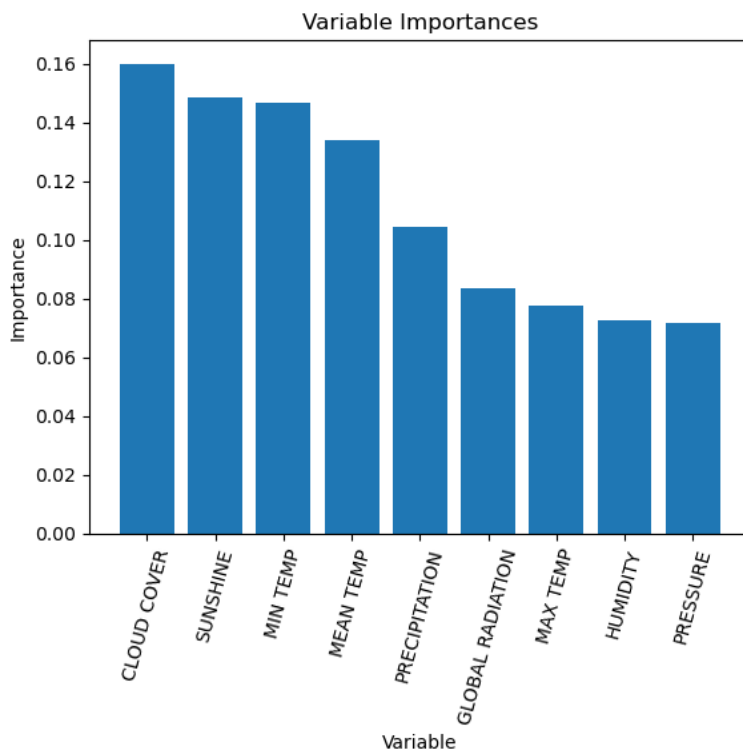
The most important variables for Kassel were **global radiation** and **precipitation**.

Because of the unusual nature of Sonnblick potentially skewing the results of the classifier, I decided to rerun the model for all stations between 2010 and 2020, minus the data from Sonnblick.

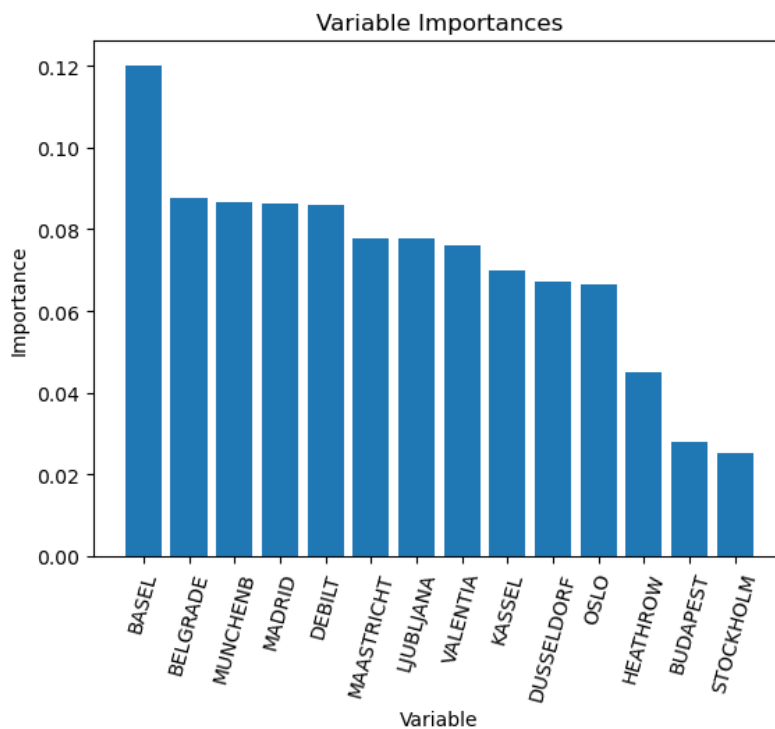
Random Forest Classification for All Stations Except Sonnblick, 2010-2020



With `n_estimators = 100`, this model has an accuracy of 54.8%



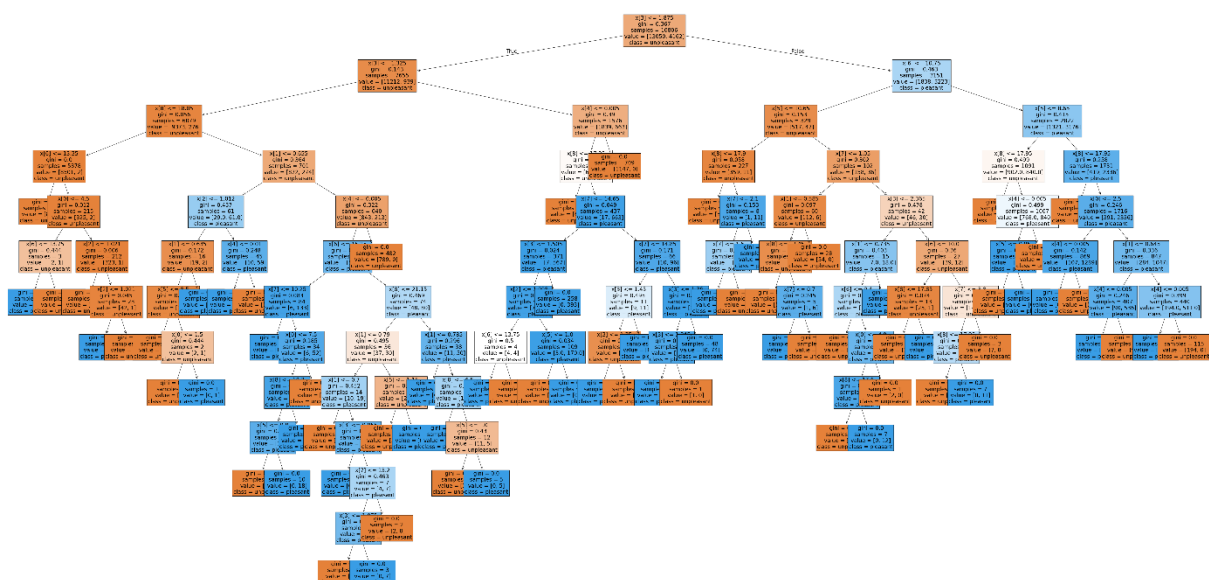
For all stations except Sonnblick between 2010 and 2020, the most important features in the weather data are **cloud cover**, **sunshine** and **minimum temperature**, while **mean temperature** also has a considerable effect.



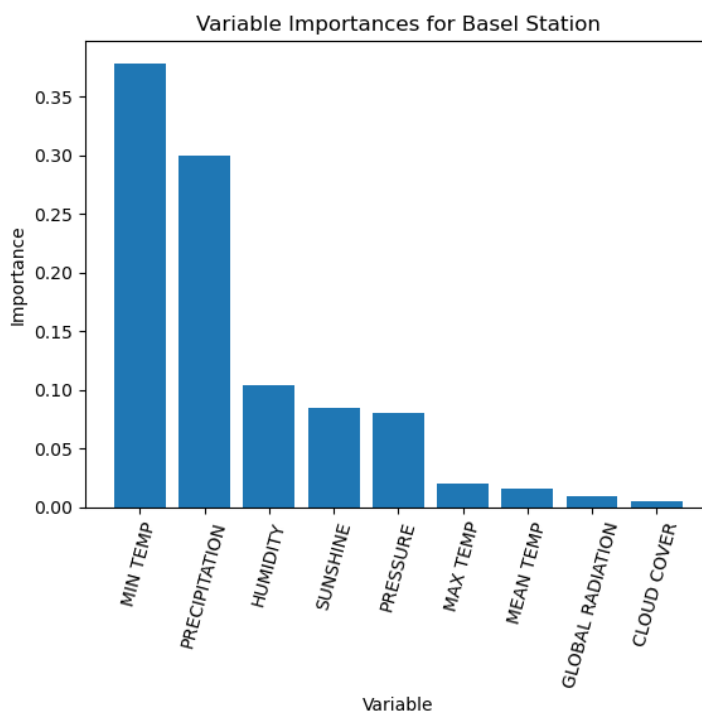
With Sonnblick removed from the weather data, the importance of the stations changes significantly. Now Basel is the station with the most feature importance by far, while Stockholm is now the least important.

It would be therefore worth looking at the classification tree and feature importances for Basel

Weather Data and Pleasant Weather Answers for Basel, 1960-2022



With $n_estimators = 100$, this model shows 100% accuracy.



Minimum temperature and **precipitation** played the biggest roles in determining the classification decisions of the random forest algorithm.

Observations

- Minimum and mean temperatures have a significant influence on the classifications made by the random forest across all stations and time periods.
- Conditions related to sunshine (sunshine and global radiation) and rain (precipitation and cloud cover) also played an important role in these classifications.
- The data from Sonnblick station had the most significant influence on the classifications by the model. However, I was concerned that the unusual nature of this station's pleasant weather data – containing exclusively negative answers – would skew the rest of the classifications and decided to rerun the model without Sonnblick data. The random forest classifier showed a similar level of accuracy, but the feature significance changed drastically in many ways: for example, the data from Stockholm now showed the lowest level of importance, compared with it having the second-highest level within the full set of data.

Next Steps

Based on these observations, pleasant weather predictions are largely dependent on weather conditions related to temperature (minimum and mean temperature), sunshine (sunshine, cloud cover, global radiation) and precipitation. Efforts should be made to maintain and update the appropriate measurement instruments for these conditions across all weather stations, and to ensure that these instruments are used by adequately-trained professionals.

It would also be worth examining the data from Sonnblick and determining, if possible, the reason for that station showing entirely negative answers in the Pleasant Weather dataset, such as year-round rain or cold temperatures, or other unique weather conditions not experienced in any of the other stations. If there is no apparent link between weather conditions and the negative answers, it may be appropriate to simply drop Sonnblick from the data when making pleasant weather-based classifications moving forward.